

About Me

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Research direction

Style Transfer, Image Editing, Computer Vision



About Me

I bring 7 publications, including 2 top-tier conference papers, plus hands-on industry research experience.



Research output

7 papers in total, including 2 top-tier conference papers

CVPR 2025

AAAI 2025

ECCV Submit

NeurIPS Submit



Industry research experience

Tencent Qingyun Talent Program, plus research internships at Alibaba, Baidu, JD.com, and Kunlun Tech



JD.com



Kunlun

Note: **Tencent Qingyun Talent Program** is Tencent's highly selective track for outstanding student researchers and engineers.

Research Overview



SigStyle



OmniStyle



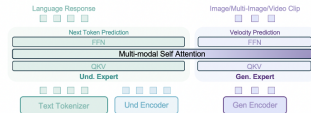
OmniStyle2



SVG



Font



UMM



CVPR 2025

OmniStyle

Filtering High-Quality Style Transfer Data at Scale

Motivation

WHY THIS PROBLEM MATTERS The paper starts from three limits in previous style transfer methods.

01

Limited generalization

Hard to handle truly arbitrary, fine-grained styles.

02

Weak controllability

Unsupervised stylization can introduce artifacts or distortion.

03

Low efficiency

Optimization or inversion makes deployment expensive.

Key question: can we build a controllable and efficient end-to-end style transfer system by learning from better paired data?

Our Solution

DATA-CENTRIC PIPELINE Our answer was to turn style transfer into a supervised learning pipeline built on filtered paired data.

01

Construct data

Build **OmniStyle-1M**, a paired dataset for stylization learning.



02

Filter data quality

Use **OmniFilter** to keep the best triplets with task-specific signals.



03

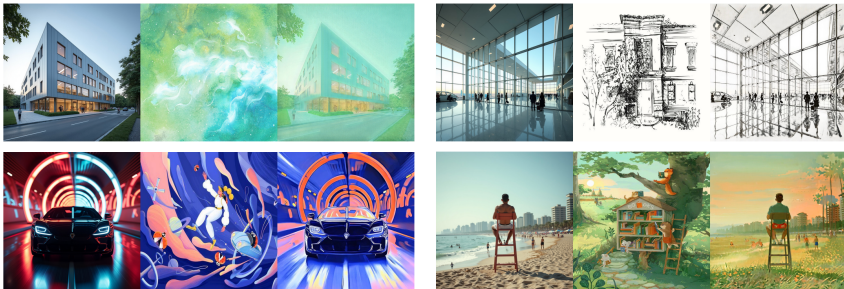
Train the model

Train **OmniStyle**, one feed-forward DiT for two stylization modes.

Bridge: the next seven slides unpack data creation first, then OmniFilter across four views, then training.

Step 1: OmniStyle-1M Overview

PIPELINE STEP 1 OmniStyle-1M is a million-scale triplet dataset with broad style coverage and diverse content.



1M+
triplets

content, style, stylized result

1,000
style categories

fine-grained style coverage

20
content categories

broad content diversity

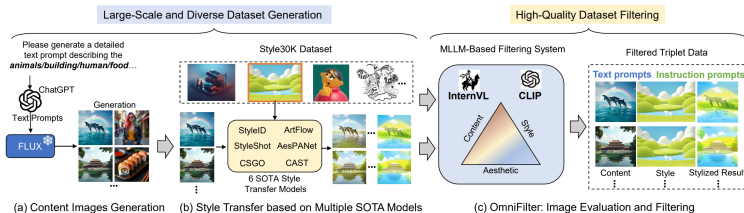
Step 1: How We Built OmniStyle-1M

PIPELINE STEP 1 We generate content images, pair them with style references, and create multiple stylized candidates per pair.

- ▶ Generate content images with FLUX across 20 content categories.
- ▶ Collect style reference images from Style30K across 1,000 style categories.
- ▶ Form content-style pairs and run six existing style transfer models.
- ▶ Keep all candidates at this stage; filtering happens in the next step.

Pipeline example

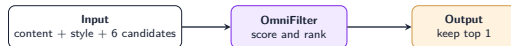
The full pairing and candidate-generation workflow is illustrated below.



Step 2: OmniFilter Overview

PIPELINE STEP 2 OmniFilter is the filtering step that turns multiple raw stylizations into one training example.

- ▶ **Input:** 1 content image, 1 style reference, and 6 generated stylizations.
- ▶ **Process:** score each candidate with three dimension checks.
- ▶ **Output:** keep the best candidate and discard the other five.



Three dimensions

Content: preserve subject and layout

Style: match the target reference

Aesthetic: keep the result visually strong

This gives us a cleaner dataset for training OmniStyle.

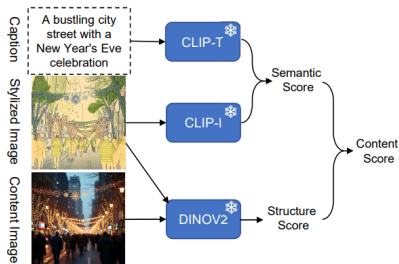
Step 2A: Content Preservation

PIPELINE STEP 2 / SIGNAL 1 The first signal checks whether stylization keeps the original subjects, semantics, and spatial structure.

- ▶ Content preservation combines semantic consistency and structural integrity.
- ▶ Semantic consistency: compare the stylized image with the content caption using CLIP image-text similarity.
- ▶ Structural integrity: compare content and stylized image embeddings with DINOv2.
- ▶ The final score gives equal weight to both parts, so the output must keep both meaning and layout.

$$C_{score} = 0.5 S_{semantic} + 0.5 S_{structural}$$

Content score pipeline



CLIP measures semantic alignment,
and DINOv2 measures structural
preservation.

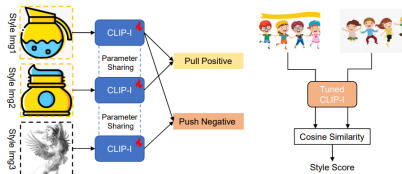
Step 2B: Style Consistency

PIPELINE STEP 2 / SIGNAL 2 The second signal checks whether the output really matches the target reference style, not just a generic artistic look.

- ▶ Style is subjective, so hand-crafted losses are not enough for reliable filtering.
- ▶ We use Style30K as supervision: same-style images are positive pairs and different-style images are negative pairs.
- ▶ Contrastive learning fine-tunes a CLIP image encoder so same styles cluster together in the embedding space.
- ▶ The similarity between the style reference and the stylized output becomes the style consistency score.

Score by embedding similarity between the target style image and the generated result.

Style score pipeline



Positive pairs are pulled together, negative pairs are pushed apart, and cosine similarity gives the final style score.

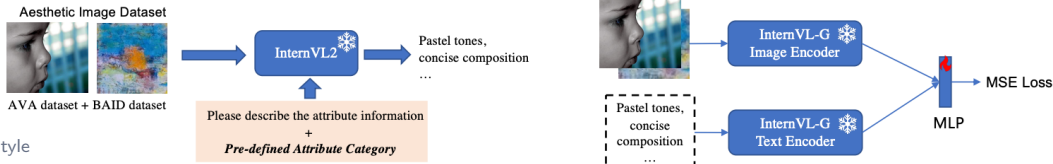
Step 2C: Aesthetic Appeal

PIPELINE STEP 2 / SIGNAL 3 The third signal rewards outputs that are not only correct, but also visually compelling.

- ▶ Many style-transfer metrics ignore visual quality, so OmniFilter adds a dedicated aesthetic scorer.
- ▶ We define 40 visual attributes, including composition, balance, color harmony, lighting, contrast, and saturation.
- ▶ InternVL2 generates attribute captions, and we combine caption features with image features.
- ▶ An MLP predicts the final score, trained on AVA first and then fine-tuned on BAID to adapt to artistic images.

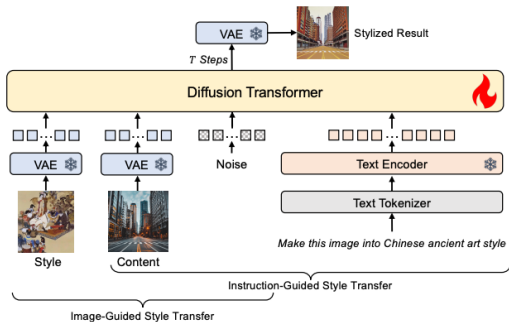
Attribute-based scoring

InternVL2 extracts attribute text, InternVL-G encodes both modalities, and an MLP regresses the final aesthetic score.



Step 3: Train the Model (OmniStyle)

PIPELINE STEP 3 A single DiT-based model handles both text-guided and image-guided stylization.



- ▶ Built on FLUX-dev / MM-DiT.
- ▶ VAEs encode content and style inputs.
- ▶ Image and text tokens join noisy latents.
- ▶ Only the diffusion transformer is fine-tuned; the other modules stay frozen.

QUANTITATIVE EVALUATION

OmniStyle leads on style consistency while staying competitive on content preservation and aesthetic quality.

Table 2. Quantitative comparison with respect to the SOTA methods on instruction-guided style transfer.

Method	Content Preservation $\uparrow$	Style Consistency $\uparrow$	Aesthetic Appeal $\uparrow$	Style Loss $\downarrow$
InstructPix2Pix [4]	0.4848	0.4965	5.7446	0.3829
HQ-Edit [14]	0.4508	0.2254	5.7777	0.3108
UltraEdit [49]	0.4906	0.5087	5.6351	0.3432
OmniGen [49]	0.4918	0.5487	5.6717	0.6272
DiffStyler [12]	0.4816	0.5127	5.4551	0.4256
OmniStyle	0.5128	0.6441	5.7512	0.2873

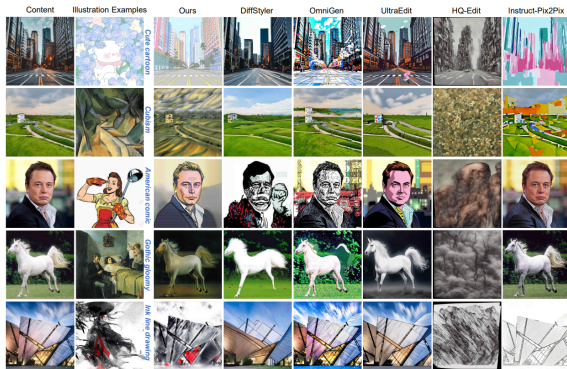
Table 3. Quantitative comparison with respect to the SOTA methods for the image-guided style transfer.

Method	Content Preservation $\uparrow$	Style Consistency $\uparrow$	Aesthetic Appeal $\uparrow$	Style Loss $\downarrow$
ArtFlow [3]	0.5184	0.6904	5.1220	0.1289
AesPANet [11]	0.5665	0.6818	5.2117	0.1128
CAST [45]	0.6165	0.6770	5.4561	0.1428
StyleID [6]	0.5121	0.6564	5.3531	0.1315
CSGO [39]	0.5067	0.7251	5.7712	0.3727
StyleShot [17]	0.5410	0.7347	5.7818	0.1500
ArtBank [48]	0.3490	0.7163	5.4980	0.1208
OmniStyle	0.5450	0.7483	5.7913	0.1086

Left: instruction-guided comparison. Right: image-guided comparison.

Qualitative EVALUATION

INSTRUCTION-GUIDED RESULTS The model preserves identity and layout while applying fine-grained styles in the instruction-guided setting.



Instruction-guided comparisons highlight stronger content preservation with more stable stylization across varied references.

Qualitative EVALUATION

IMAGE-GUIDED RESULTS The same model transfers color, texture, and composition reliably from a single style reference image.



Reference-image comparisons show closer style matching while keeping the original scene structure intact.

Summary

KEY TAKEAWAYS The paper can be summarized as a data pipeline: **build the dataset, filter quality, then train one unified model.**

- ▶ **Dataset:** OmniStyle-1M constructs large-scale triplets of content images, style references, and stylized results, giving the model broad supervision across diverse styles and scenes.
- ▶ **Filter:** OmniFilter scores candidates on content preservation, style consistency, and aesthetic appeal, then keeps the best examples for training.
- ▶ **Model:** OmniStyle is a unified DiT-based feed-forward model that supports both instruction-guided and image-guided stylization with strong controllability.